GATE CS — Engineering Mathematics

Topic-wise past questions (2016–2026) + new practice questions with detailed answers

Prepared for GATE CS 2027 from the papers in pAPER/ of the GATE2027 repo.

Topic	Past questions	Total marks	New practice Qs
Probability & Statistics	30	49	2
Linear Algebra	28	42	2
Sets, Relations, Functions, Groups, Lattices	23	35	2
Graph Theory	22	35	2
Combinatorics, Recurrences, Generating Functions	16	23	2
Propositional & First-Order Logic	14	19	2
Calculus	17	19	2
Total	**150**	**222**	**14**

How to read this

- Each topic has two parts: **A. Past-paper questions** (newest first, tagged year · set · question number · marks) and **B. New practice questions** with step-by-step solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Math symbols and figures may be lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original PDF in pAPER/<year>/ for the exact wording and any figure.
- New practice questions are original, written in GATE style; numerical answers were verified.

Probability & Statistics

3.3 marks per paper · 30 questions · appeared in 14 of 15 papers. Conditional probability/Bayes, expectation and variance, PDFs, dice/coin/urn problems.

A. Past-paper questions (30)

2026 CS1 · Q.11 · 1 mark
An urn contains one red ball and one blue ball. At each step, a ball is picked uniformly at random from the urn, and this ball together with another ball of the same color is put back in the urn. The probability that there are equal number of red and blue balls after two steps is
(A) 1/4
(B) 1/3
(C) 1/2
(D) 2/3

2026 CS1 · Q.58 · 2 marks
Let X be a random variable which takes values in the set {1, 2, 3, 4, 5, 6, 7, 8}. 1 Further, Pr(X = 1) = Pr(X = 2) = Pr(X = 5) = Pr(X = 7) = 6 and 1 Pr(X = 3) = Pr(X = 4) = Pr(X = 6) = Pr(X = 8) = 12 . The expected value of X, denoted by E[X], is equal to _____. (rounded off to two decimal places)

2026 CS2 · Q.14 · 1 mark
The probability density function f(x) of a random variable X which takes real values is 1 x2 f(x) = exp(−), x ∈ (−∞, +∞) 3√2π 18 Which one of the following statements is correct about the random variable X ?
(A) X is an exponential random variable
(B) X is a normal random variable
(C) X is a Poisson random variable
(D) X is a uniform random variable

2026 CS2 · Q.63 · 2 marks
Suppose an unbiased coin is tossed 6 times. Each coin toss is independent of all previous coin tosses. Let E1 be the event that among the second, fourth, and sixth coin tosses, there are at least two heads. Let E2 be the event that among the first, second, third, and fifth coin tosses, there are equal number of heads and tails. The conditional probability P(E1 | E2) is equal to _____. (rounded off to one decimal place)

2025 CS1 · Q.32 · 1 mark
A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability P(head) = 0.5 and for a fake coin, P(head) = 1. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (rounded off to two decimal places)

2025 CS1 · Q.56 · 2 marks
Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____ . (rounded off to three decimal places)

2025 CS1 · Q.58 · 2 marks
Consider a probability distribution given by the density function P(x). Cx 2 , for 1 ≤ x ≤ 4 P(x) = { 0, for x < 1 or x > 4 The probability that x lies between 2 and 3, i.e., P(2 ≤ x ≤ 3) is _____. (rounded off to three decimal places)

2025 CS2 · Q.64 · 2 marks
A quadratic polynomial (x − α)(x − β) over complex numbers is said to be square invariant if (x − α)(x − β) = (x − α 2)(x − β 2). Suppose from the set of all square invariant quadratic polynomials we choose one at random. The probability that the roots of the chosen polynomial are equal is _____. (rounded off to one decimal place)

2025 CS2 · Q.65 · 2 marks
The unit interval (0,1) is divided at a point chosen uniformly distributed over (0,1) in R into two disjoint subintervals. The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)

2024 CS1 · Q.14 · 1 mark
Consider a permutation sampled uniformly at random from the set of all permutations of {1, 2, 3, ⋯ , n} for some n ≥ 4. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?
(A) The events X and Y are mutually exclusive
(B) The events X and Y are independent
(C) Either event X or Y must occur
(D) Event X is more likely than event Y

2024 CS1 · Q.27 · 1 mark
Let A and B be two events in a probability space with P
(A) = 0.3, P
(B) = 0.5, and P(A ∩ B) = 0.1. Which of the following statements is/are TRUE?
(A) The two events A and B are independent
(B) P(A ∪ B) = 0.7
(C) P(A ∩ B c) = 0.2, where B c is the complement of the event B P(Ac ∩ B c) = 0.4, where Ac and B c are the complements of the events A and B,
(D) respectively

2024 CS1 · Q.63 · 2 marks
A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____

2024 CS2 · Q.18 · 1 mark
When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (i.e., 1, 2, 3, 4, 5, and 6) is 1
(A) 324 5
(B) 324 7
(C) 324 11
(D) 324

2024 CS2 · Q.44 · 2 marks
Let x and y be random variables, not necessarily independent, that take real values in the interval [0,1]. Let z = xy and let the mean values of x, y, z be $\bar{x}$, $\bar{y}$, $\bar{z}$, respectively. Which one of the following statements is TRUE?
(A) $\bar{z} = \bar{x}\,\bar{y}$
(B) $\bar{z} \leq \bar{x}\,\bar{y}$
(C) $\bar{z} \geq \bar{x}\,\bar{y}$
(D) $\bar{z} \leq \bar{x}$

2023 · Q.53 · 2 marks
Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?
(A) A and B are independent.
(B) A and C are independent.
(C) B and C are independent.
(D) Prob(B|C) = Prob
(B) CS

2021 · Q.18 · 1 mark [OCR]
The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2. For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is

2021 · Q.35 · 2 marks [OCR]
Consider the two statements. Sy: There exist random variables X and Y such that 2 (E[(x -E(x)) (vy -E(Y))]) > var{x] var(¥] Sq: For all random variables X and Y, cov[X,¥] = E[|X —E[X]| |Y -E(Y]]|] Which one of the following choices is correct?
(A) | Both S; and S2 are true.
(B) | S, is true, but Sy is false.
(C) | Sj is false, but S2 is true.
(D) | Both S; and Sy are false. CS, Set-1 - Copyright © GATE 2021

2021 · Q.54 · 2 marks [OCR]
A sender (S) transmits a signal, which can be one of the two kinds: H and L with probabilities 0.1 and 0.9 respectively, to a receiver (R). In the graph below, the weight of edge (u,v) is the probability of receiving v when u is transmitted, where u,v € {H, L}. For example, the probability that the received signal is L given the transmitted signal was H, is 0.7. H os H 0.8 0.7 L 02 L If the received signal is H, the probability that the transmitted signal was H (rounded to 2 decimal places) is

2020 · Q.45 · 2 marks [OCR]
For n> 2,let a € {0,1}" be a non-zero vector. Suppose that x is chosen uniformly at random from {0,1}". Then, the probability that D7, a;x; is an odd number is

2019 · Q.22 · 1 mark [OCR]
Two numbers are chosen independently and uniformly at random from the set {1,2,..., 13}. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) ‘binary representations have the same most significant bit is

2019 · Q.46 · 2 marks [OCR]
Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.) Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places)

2019 · Q.47 · 2 marks [OCR]
Suppose Y is distributed uniformly in the open interval (1,6). The probability that the polynomial 3x? + 6x¥ + 3Y + 6 has only real roots is (rounded off to 1 decimal place)

2018 · Q.15 · 1 mark
Two people, P and Q, decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____.

2018 · Q.44 · 2 marks
Consider Guwahati (G) and Delhi
(D) whose temperatures can be classified as high (H), medium (M) and low (L). Let P(HG) denote the probability that Guwahati has high temperature. Similarly, P(MG) and P(LG) denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use P(HD), P(MD) and P(LD) for Delhi. The following table gives the conditional probabilities for Delhi’s temperature given Guwahati’s temperature. HD MD LD HG 0.40 0.48 0.12 MG 0.10 0.65 0.25 LG 0.01 0.50 0.49 Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (HG) then the probability of Delhi also having a high temperature (HD) is 0.40; i.e., P(HD |HG) = 0.40. Similarly, the next two entries are P(MD |HG) = 0.48 and P(LD |HG) = 0.12. Similarly for the other rows. If it is known that P(HG) = 0.2, P(MG) = 0.5, and P(LG) = 0.3, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

2017 CS1 · Q.19 · 1 mark [OCR]
r . . . - D | r , Let Y be a Gaussian random variable with mean 0 and variance o~. Let Y =max(X.0) where max(a.b) is the maximum of a and D . The median of Y is

2017 CS2 · Q.26 · 2 marks [OCR]
1 P and Q are considering to apply for a job. The probability that P applies for the job is 3? the 1 probability that P applies for the job given that Q applies for the job is 3 and the probability that Q 1 applies for the job given that P applies for the job is = Then the probability that P does not apply for the job given that Q does not apply for the job is ol ~~ a py As
(B) i. . O) 7G

2017 CS2 · Q.31 · 2 marks [OCR]
For any discrete random variable X . with probability mass function N P(X = j)=p,. p, 29. 7 €{0.....N}, and » p, =1. define the polynomial function j=0 i £-@)= >» Pp,’ . For a certain discrete random variable Y , there exists a scalar 8 €[0,1] such j=0 that g,(z)=(1-B +6 z)TM . The expectation of Y is
(A) NB (-8)
(B) NB
(C) N(1-8)
(D) Not expressible in terms of N and 8 alone

2017 CS2 · Q.48 · 2 marks [OCR]
If a random variable X has a Poisson distribution with mean 5, then the expectation E[(X + 2)?] equals

2016 · Q.4 · 1 mark
A probability density function on the interval [a, 1] is given by 1/x2 and outside this interval the value of the function is zero. The value of a is . √

2016 · Q.29 · 2 marks
Consider the following experiment. Step 1. Flip a fair coin twice. Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop. Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop. Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1. The probability that the output of the experiment is Y is (up to two decimal places) .

B. New practice questions with detailed answers

Q1 · 1 mark · NAT
Two fair six-sided dice are rolled. The probability that the sum is a prime number, rounded to 3 decimal places, is _____.

> **Answer: 0.417**
> Prime sums possible: 2, 3, 5, 7, 11.
> Ways: sum 2 → 1, sum 3 → 2, sum 5 → 4, sum 7 → 6, sum 11 → 2. Total favourable = 1+2+4+6+2 = 15.
> P = 15/36 = 0.41666... ≈ **0.417**.
> *Tip:* list sums by number of ways (1,2,3,4,5,6,5,4,3,2,1 for sums 2...12) so you never recount.

Q2 · 2 marks · MCQ
Urn A holds 3 red and 2 blue balls. Urn B holds 1 red and 4 blue balls. An urn is chosen with equal probability and one ball is drawn from it. The ball is red. The probability that it came from urn A is
(A) 1/2 (B) 3/5 (C) 3/4 (D) 4/5

> **Answer: (C) 3/4**
> Bayes' theorem: P(A|red) = P(red|A)·P(A) / [P(red|A)·P(A) + P(red|B)·P(B)].
> P(red|A) = 3/5, P(red|B) = 1/5, P(A) = P(B) = 1/2.
> Numerator = ½·3/5 = 0.3. Denominator = 0.3 + ½·1/5 = 0.3 + 0.1 = 0.4.
> P = 0.3/0.4 = **3/4**.

Linear Algebra

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A. Past-paper questions (28)

2026 CS1 · Q.13 · 1 mark
For n > 1, the maximum multiplicity of any eigenvalue of an n × n matrix with elements from R is
(A) n
(B) n−1
(C) 1
(D) n+1

2026 CS1 · Q.20 · 1 mark
Let n > 1. Consider an n × n matrix M with its elements from R. Let the vector (0, 1, 0, 0, ... , 0) ∈ Rn be in the null space of M. Which of the following options is/are always correct?
(A) Determinant of M is 1
(B) Determinant of M is 0
(C) Rank of M is 1
(D) There are at least two non-zero vectors in the null space of M

2026 CS2 · Q.31 · 1 mark
Consider the system of linear equations given below. ax + y = b 16x + ay = 24 Suppose the values of a and b are chosen such that the system of linear equations produce multiple solutions. Then the product of a and b is _____. (answer in integer)

2026 CS2 · Q.62 · 2 marks
The determinant of a 4 × 4 matrix A is 3. The value of the determinant of 2A is _____. (answer in integer)

2025 CS1 · Q.23 · 1 mark
real-valued constant. Which of the following option(s) is/are CORRECT? x + ky = 1 kx + y = −1
(A) There is exactly one value of k for which the above system of equations has no solution.
(B) There exist an infinite number of values of k for which the system of equations has no solution.
(C) There exists exactly one value of k for which the system of equations has exactly one solution.
(D) There exists exactly one value of k for which the system of equations has an infinite number of solutions.

2025 CS1 · Q.41 · 2 marks
1 1 A=[ ] 1 −1 What are the eigenvalues of the matrix A13 ?
(A) 1, −1
(B) 2√2, −2√2
(C) 4√2, −4√2
(D) 64√2, −64√2

2025 CS2 · Q.11 · 1 mark
1 2 If A = ( ), then which ONE of the following is A8 ? 2 −1
(A) 25 0 ( ) 0 25
(B) 125 0 ( ) 0 125
(C) 625 0 ( ) 0 625
(D) 3125 0 ( ) 0 3125 x 1

2025 CS2 · Q.14 · 1 mark
Let L, M, and N be non-singular matrices of order 3 satisfying the equations L2 = L−1 , M = L8 and N = L2 . Which ONE of the following is the value of the determinant of (M − N)?
(A) 0
(B) 1
(C) 2
(D) 3

2025 CS2 · Q.44 · 2 marks
Consider a system of linear equations PX = Q where P ∈ R3×3 and Q ∈ R3×1 . Suppose P has an LU decomposition, P = LU, where 1 0 0 u11 u12 u13 l L = [21 1 0] and U = [0 u22 u23]. l31 l32 1 0 0 u33 Which of the following statement(s) is/are TRUE?
(A) The system PX = Q can be solved by first solving LY = Q and then UX = Y.
(B) If P is invertible, then both L and U are invertible.
(C) If P is singular, then at least one of the diagonal elements of U is zero.
(D) If P is symmetric, then both L and U are symmetric.

2024 CS1 · Q.12 · 1 mark
The product of all eigenvalues of the matrix [4 5 6] is 7 8 9
(A) −1
(B) 0
(C) 1
(D) 2

2024 CS1 · Q.49 · 2 marks
Let A be any n × m matrix, where m > n. Which of the following statements is/are TRUE about the system of linear equations Ax = 0 ?
(A) There exist at least m − n linearly independent solutions to this system There exist m − n linearly independent vectors such that every solution is a linear
(B) combination of these vectors
(C) There exists a non-zero solution in which at least m − n variables are 0
(D) There exists a solution in which at least n variables are non-zero

2024 CS2 · Q.47 · 2 marks
obtained from A by swapping two rows. Which of the following statements is/are TRUE?
(A) The determinant of B is the negative of the determinant of A
(B) If A is invertible, then B is also invertible
(C) If Ais symmetric, then B is also symmetric
(D) If the trace of A is zero, then the trace of Bis also zero

2023 · Q.18 · 1 mark
Let ⎡ ⎤ 1 2 3 4 ⎢4 1 2 3⎥⎥ A=⎢ ⎣3 4 1 2⎦ 2 3 4 1 and ⎡ ⎤ 3 4 1 2 ⎢4 1 2 3⎥⎥. B=⎢ ⎣1 2 3 4⎦ 2 3 4 1 Let det
(A) and det
(B) denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?
(A) det
(A) = det
(B)
(B) det
(B) = −det
(A)
(C) det
(A) = 0
(D) det(AB) = det
(A) + det
(B) CS

2022 · Q.20 · 1 mark
Consider the following two statements with respect to the matrices Amn , Bnm , Cnn and Dnn . Statement 1: tr(AB) = tr(BA) Statement 2: tr(CD) = tr(DC) where tr() represents the trace of a matrix. Which one of the following holds?
(A) Statement 1 is correct and Statement 2 is wrong.
(B) Statement 1 is wrong and Statement 2 is correct.
(C) Both Statement 1 and Statement 2 are correct.
(D) Both Statement 1 and Statement 2 are wrong.

2022 · Q.45 · 2 marks
Consider solving the following system of simultaneous equations using LU decomposition. x1 x2 2 x3 4 x1 3 x2 x3 7 2 x1 x2 5 x3 7 where L and U are denoted as L11 0 0 U11 U12 U13 L L21 L22 0 , U 0 U 22 U 23 L L33 0 U 33 31 L32 0 Which one of the following is the correct combination of values for L32 , U 33 , and x1 ?
(A) 1 L32 2, U 33 , x1 1 2
(B) L32 2, U 33 2, x1 1
(C) 1 L32 , U 33 2, x1 0 2
(D) 1 1 L32 , U 33 , x1 0 2 2

2022 · Q.53 · 2 marks
Which of the following is/are the eigenvector(s) for the matrix given below? 9 6 2 4 8 6 3 1 20 15 8 5 32 21 7 12
(A) 1 1 0 1
(B) 1 0 1 0
(C) 1 0 2 2
(D) 0 1 3 0

2021 · Q.52 · 2 marks [OCR]
Consider the following matrix. o1i1i1 1011 1101 1110 The largest eigenvalue of the above matrix is

2020 · Q.27 · 2 marks [OCR]
Let A and B be two n X n matrices over real numbers. Let rank(M) and det(M) denote the rank and determinant of a matrix M, respectively. Consider the following statements. I. rank(AB) = rank
(A) rank
(B) II. det(AB) = det
(A) det
(B) III. rank(A + B) Ss rank
(A) + rank
(B) IV. det(A + B) < det
(A) + det
(B) Which of the above statements are TRUE?
(A) land II only (8) Land IV only (©) IL and II only (0) III and IV only

2019 · Q.9 · 1 mark [OCR]
Let X be a square matrix. Consider the following two statements on X. L X is invertible. I. Determinant of X is non-zero. Which one of the following is TRUE?
(A) Limplies II; I does not imply I.
(B) I implies I; I does not imply IL.
(C) I does not imply I; II does not imply I.
(D) I and II are equivalent statements.

2019 · Q.44 · 2 marks [OCR]
Consider the following matrix: 124 8 pe|1 3 9 27 14 16 64 1 5 25 125 The absolute value of the product of Eigen values of R is

2018 · Q.17 · 1 mark
Consider a matrix A uv T where u 1 , v 1 . Note that vT denotes the 2 1 transpose of v. The largest eigenvalue of A is _____.

2018 · Q.26 · 2 marks
1 Consider a matrix P whose only eigenvectors are the multiples of . 4 Consider the following statements. (I) P does not have an inverse (II) P has a repeated eigenvalue (III) P cannot be diagonalized Which one of the following options is correct?
(A) Only I and III are necessarily true
(B) Only II is necessarily true
(C) Only I and II are necessarily true
(D) Only II and III are necessarily true CS 7/20 GATE 2018

2017 CS1 · Q.3 · 1 mark [OCR]
n . n Let ¢,.....¢, be scalars. not all zero. such that %, c,a; = 9 where a, are column vectors inR’. i=I Consider the set of linear equations Ax=b n where 4=[a,.....a,] and b= >» a; . The set of equations has i=I
(A) a unique solution at x = J, where J, denotes a 7 -dimensional vector of all 1
(B) no solution
(C) infinitely many solutions
(D) finitely many solutions

2017 CS1 · Q.30 · 2 marks [OCR]
. 2 . ; . Let WU and Vv be two vectors in R~ whose Euclidean norms satisfy | u \|= 2 | Vv | . What is the value of @ such that w=" + av bisects the angle between and v?
(A) 2 @®) 1/2
(C) 1
(D) -1/2

2017 CS1 · Q.31 · 2 marks [OCR]
non 2 a Let A be 7x1 real valued square symmetric matrix of rank 2 with 2 >, Ay =50. Consider the i=1 j=I following statements. (I) One eigenvalue must be in[—5. 5] (II) The eigenvalue with the largest magnitude must be strictly greater than 5 Which of the above statements about eigenvalues of 4 is/are necessarily CORRECT?
(A) — Both (D and (ID
(B) (only
(C) (only
(D) Neither (I) nor (II) =====PAGE 15=====

2017 CS2 · Q.22 · 1 mark [OCR]
1 1 -1 —1 -2 —1 LetP=|2 -—3 4]andQ=]6 12. 6 |betwo matrices. 3 —2 3 5 .Jt0 6 Then the rank of P + Q is

2017 CS2 · Q.52 · 2 marks [OCR]
If the characteristic polynomial of a 3 x 3 matrix M over R (the set of real numbers) is 23 — 42? + aA +30. aE RB, and one eigenvalue of M is 2. then the largest among the absolute values of the eigenvalues of M is

2016 · Q.5 · 1 mark
Two eigenvalues of a 3 × 3 real matrix P are (2 + −1) and 3. The determinant of P is .

B. New practice questions with detailed answers

Q1 · 1 mark · NAT
A 3 × 3 real matrix A has eigenvalues 1, 2 and 4. The trace of A^2 is _____.

Answer: 21
If λ is an eigenvalue of A, then λ^2 is an eigenvalue of A^2. So A^2 has eigenvalues 1, 4, 16.
Trace = sum of eigenvalues = 1 + 4 + 16 = **21**.
Rule: trace = Σλ, det = Πλ, and f(A) has eigenvalues f(λ).

Q2 · 2 marks · MSQ
Consider the system x + y + z = 3, x + 2y + 3z = 6, 2x + 3y + kz = 9. Which statements are TRUE?
(A) For k = 4 the system has infinitely many solutions.
(B) For k ≠ 4 the system has the unique solution (x, y, z) = (0, 3, 0).
(C) For some value of k the system is inconsistent.
(D) The determinant of the coefficient matrix equals k − 4.

Answer: (A), (B), (D)
Eq2 − Eq1: y + 2z = 3. Eq3 − 2·Eq1: y + (k−2)z = 3. Subtract: (k − 4)z = 0.
• k ≠ 4 → z = 0, then y = 3, x = 0. Unique solution (0, 3, 0) → (B) true.
• k = 4 → the equation becomes 0 = 0, rank 2 < 3 unknowns, consistent → infinitely many solutions → (A) true.
• The right side of the reduced equation is always 0, so the system is never inconsistent → (C) false.
• det = 1(2k−9) − 1(k−6) + 1(3−4) = 2k − 9 − k + 6 − 1 = k − 4 → (D) true.

Sets, Relations, Functions, Groups, Lattices

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A. Past-paper questions (23)

2026 CS2 · Q.26 · 1 mark
Let R be a binary relation on the set {1, 2, ... , 10}, where (x, y) ∈ R if the product of x and y is square of an integer. Which of the following properties is/are satisfied by R?
(A) Reflexive
(B) Symmetric
(C) Transitive
(D) Antisymmetric

2026 CS2 · Q.64 · 2 marks
Consider a function f: (0,1) → {0, 1} defined as follows. For a real number r ∈ (0,1) , f(r) = 1 if the second digit after the decimal point in r is one of the four digits 2, 3, 6 and 7. Otherwise, f(r) is equal to 0. The number of points in (0,1) at which f is discontinuous is _____. (answer in integer)

2025 CS1 · Q.17 · 1 mark
g(.) is a function from A to B, f(.) is a function from B to C, and their composition defined as f(g(.)) is a mapping from A to C. If f(.) and f(g(.)) are onto (surjective) functions, which ONE of the following is TRUE about the function g(.)?
(A) g(.) must be an onto (surjective) function.
(B) g(.) must be a one-to-one (injective) function.
(C) g(.) must be a bijective function, that is, both one-to-one and onto.
(D) g(.) is not required to be a one-to-one or onto function.

2025 CS1 · Q.49 · 2 marks
A = {0, 1, 2, 3, ... } is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, f1 , f2 ∈ F, we define (f1 ⊙f2)(n) = f1 (n) + f2 (n) for every number n in A. Which of the following is/are CORRECT about the mathematical structure (F, ⊙)?
(A) (F, ⊙) is an Abelian group.
(B) (F, ⊙) is an Abelian monoid.
(C) (F, ⊙) is a non-Abelian group.
(D) (F, ⊙) is a non-Abelian monoid.

2025 CS2 · Q.42 · 2 marks
Let F be the set of all functions from {1, ... , n} to {0,1}. Define the binary relation ≼ on F as follows: ∀f, g ∈ F, f ≼ g if and only if ∀x ∈ {1, ... , n}, f(x) ≤ g(x), where 0 ≤ 1. Which of the following statement(s) is/are TRUE?
(A) ≼ is a symmetric relation
(B) (F, ≼) is a partial order
(C) (F, ≼) is a lattice
(D) ≼ is an equivalence relation

2024 CS1 · Q.32 · 1 mark
Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from A × A to A ∪ B. The number of possible values of |A| is _____

2024 CS1 · Q.52 · 2 marks
Consider the operators ◊ and □ defined by a ◊ b = a + 2b, a□b = ab, for positive integers. Which of the following statements is/are TRUE?
(A) Operator ◊ obeys the associative law
(B) Operator □ obeys the associative law
(C) Operator ◊ over the operator □ obeys the distributive law
(D) Operator □ over the operator ◊ obeys the distributive law

2024 CS2 · Q.34 · 1 mark
Let P be the partial order defined on the set {1,2,3,4} as follows P = {(x, x)|x ∈ {1,2,3,4}} ∪ {(1,2), (3,2), (3,4)} The number of total orders on {1,2,3,4} that contain P is _____

2024 CS2 · Q.63 · 2 marks
Let Zn be the group of integers {0, 1, 2, ... , n − 1} with addition modulo n as the group operation. The number of elements in the group Z2 × Z3 × Z4 that are their own inverses is _____

2023 · Q.49 · 2 marks
Let f : A → B be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation ∼ on the set A as a1 ∼ a2 if f (a1) = f (a2) , where a1 , a2 ∈ A. Let E = {[x] : x ∈ A} be the set of all the equivalence classes under ∼. Define a new mapping F : E → B as F ([x]) = f (x), for all the equivalence classes [x] in E. Which of the following statements is/are TRUE?
(A) F is NOT well-defined.
(B) F is an onto (or surjective) function.
(C) F is a one-to-one (or injective) function.
(D) F is a bijective function. CS

2023 · Q.51 · 2 marks
Let X be a set and 2X denote the powerset of X. Define a binary operation Δ on 2X as follows: AΔB = (A − B) ∪ (B − A) . Let H = (2X , Δ). Which of the following statements about H is/are correct?
(A) H is a group.
(B) Every element in H has an inverse, but H is NOT a group.
(C) For every A ∈ 2X , the inverse of A is the complement of A.
(D) For every A ∈ 2X , the inverse of A is A. CS

2022 · Q.27 · 1 mark
Which of the following statements is/are TRUE for a group G ?
(A) If for all x, y G , (xy)2 x 2 y 2 , then G is commutative.
(B) If for all x G , x 2 1, then G is commutative. Here, 1 is the identity element of G .
(C) If the order of G is 2 , then G is commutative.
(D) If G is commutative, then a subgroup of G need not be commutative.

2021 · Q.34 · 2 marks [OCR]
Let G be a group of order 6, and H be a subgroup of G such that 1 < |H| < 6. Which one of the following options is correct?
(A) | Both G and H are always cyclic.
(B) | G may not be cyclic, but H is always cyclic.
(C) | Gis always cyclic, but H may not be cyclic.
(D) | Both G and H may not be cyclic.

2021 · Q.43 · 2 marks [OCR]
A relation R is said to be circular if aRb and bRe together imply cRa. Which of the following options is/are correct?
(A) | If a relation S is reflexive and symmetric, then S is an equivalence relation.
(B) | If a relation S is circular and symmetric, then S is an equivalence relation.
(C) | Ifa relation S is reflexive and circular, then S is an equivalence relation.
(D) | Ifa relation S is transitive and circular, then S is an equivalence relation. CS, Set-1 - Copyright © GATE 2021

2020 · Q.17 · 1 mark [OCR]
Let R be the set of all binary relations on the set {1,2,3}. Suppose a relation is chosen from R at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is

2020 · Q.18 · 1 mark [OCR]
Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is

2019 · Q.10 · 1 mark [OCR]
Let G be an arbitrary group. Consider the following relations on G: R,: Va,b €G, aRyb ifand only if ag € G such that a = g bg Rz:Va,b € G, a Rob if and only ifa = b+ Which of the above is/are equivalence relation/relations?
(A) R; and Ry
(B) R, only
(C) R2 only
(D) Neither R; nor Rz cs 2/15 GATE 2019 Computer Science and Information Technology Set-2

2019 · Q.21 · 1 mark [OCR]
The value of 3°! mod5 is

2019 · Q.48 · 2 marks [OCR]
Let X be the set of all bijections from {1,...,5} to {1,...,5}, where id denotes the identity function, i.e. id(j) = j,Vj. Let © denote composition on functions. For a string x = Xy XQ°Xpy EL", n=O, let w(x) = xy 0x7 O+ O Xp. Consider the language L = {x € 2*| z(x) = id}. The minimum number of states in any DFA accepting L is cs 11s GATE 2019 Computer Science and Information Technology Set-2

2018 · Q.19 · 1 mark
Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____.

2018 · Q.27 · 2 marks
Let N be the set of natural numbers. Consider the following sets. P: Set of Rational numbers (positive and negative) Q: Set of functions from {0, 1} to N R: Set of functions from N to {0, 1} S: Set of finite subsets of N. Which of the sets above are countable?
(A) Q and S only
(B) P and S only
(C) P and R only
(D) P, Q and S only

2017 CS2 · Q.21 · 1 mark [OCR]
Consider the set XY = {a,b,c,d,e} under the partial ordering R={(a.a).(a.b).(a.c).(a.d).(a.e).(b.b). (b.c). (b. e). (c.0). (c.e). (d.d). (d.e).(e.e)} - The Hasse diagram of the partial order (Y,R) is shown below. The minimum number of ordered pairs that need to be added to R to make (YR) a lattice is

2016 · Q.28 · 2 marks
A function f : N+ → N+ , defined on the set of positive integers N+ , satisfies the following properties: f (n) = f (n/2) if n is even f (n) = f (n + 5) if n is odd Let R = {i | ∃ j : f (j) = i} be the set of distinct values that f takes. The maximum possible size of R is .

B. New practice questions with detailed answers

Q1 · 1 mark · NAT
The number of equivalence relations on the set {1, 2, 3, 4} is _____.

Answer: 15
Equivalence relations on a set correspond one-to-one with partitions of the set. The count is the Bell number B(4).
Using Stirling numbers of the second kind: S(4,1)=1, S(4,2)=7, S(4,3)=6, S(4,4)=1. Sum = **15**.
Bell numbers: 1, 1, 2, 5, 15, 52, 203...

Q2 · 2 marks · NAT
The divisors of 36 are ordered by divisibility. The number of edges in the Hasse diagram of this poset is _____.

Answer: 12
$36 = 2^2 \cdot 3^2$. Each divisor is $2^a 3^b$ with a, b ∈ {0,1,2}, so the poset is a 3 × 3 grid (product of two chains).
Covering edges raise a or b by exactly 1. Raising a: 2 choices of step × 3 values of b = 6. Raising b: 6. Total = **12**.
(Divisors: 1,2,3,4,6,9,12,18,36 — nine nodes.)

Graph Theory

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A. Past-paper questions (22)

2026 CS1 · Q.47 · 2 marks
Let G(V, E) be a simple, undirected graph. A vertex cover of G is a subset $V' \subseteq V$ such that for every $(u, v) \in E$, $u \in V'$ or $v \in V'$. Let the size of the smallest vertex cover in G be k. Let S be any vertex cover of size k. For a vertex $v \in V$, which of the following constraints will always ensure that $v \in S$?
(A) The degree of v is at least k + 1
(B) The vertex v is on a path of length k + 1
(C) The vertex v is on a cycle of length k + 1
(D) The vertex v is a part of a clique of size k

2026 CS1 · Q.55 · 2 marks
An undirected, unweighted, simple graph G(V, E) is said to be 2-colorable if there exists a function $c: V \to \{0, 1\}$ such that for every $(u, v) \in E$, $c(u) \neq c(v)$. Which of the following statements about 2-colorable graphs is/are true?
(A) If G is 2-colorable, then G may contain cycles of odd length
(B) If G is 2-colorable, then G may contain cycles of even length
(C) An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|V| + |E|)$, if G is represented as an adjacency list
(D) An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|E| \log|V|)$, if G is represented as an adjacency list

2026 CS1 · Q.57 · 2 marks
Let G be an undirected graph, which is a path on 8 vertices. The number of matchings in G is _____. (answer in integer)

2026 CS2 · Q.36 · 2 marks
Consider a complete graph Kn with n vertices (n > 4). Note that multiple spanning trees can be constructed over Kn . Each of these spanning trees is represented as a set of edges. The Jaccard coefficient between any two sets is defined as the ratio of the size of the intersection of the two sets to the size of the union of the two sets. Which one of the following options gives the lowest possible value for the Jaccard coefficient between any two spanning trees of Kn ?
(A) 1 n
(B) 1 2n − 3
(C) 0
(D) 1 n−1

2024 CS1 · Q.34 · 1 mark
The number of spanning trees in a complete graph of 4 vertices labelled A, B, C, and D is _____

2024 CS1 · Q.51 · 2 marks
The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. Let G be any graph with n vertices and chromatic number k. Which of the following statements is/are always TRUE?
(A) G contains a complete subgraph with k vertices
(B) G contains an independent set of size at least n/k
(C) G contains at least k(k − 1)/2 edges
(D) G contains a vertex of degree at least k

2024 CS2 · Q.17 · 1 mark
Let A be the adjacency matrix of a simple undirected graph G. Suppose A is its own inverse. Which one of the following statements is always TRUE?
(A) G is a cycle
(B) G is a perfect matching
(C) G is a complete graph
(D) There is no such graph G

2024 CS2 · Q.60 · 2 marks
The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. The chromatic number of the following graph is _____

2023 · Q.30 · 1 mark
Let A be the adjacency matrix of the graph with vertices {1, 2, 3, 4, 5}. 3 5 2 1 4 Let $\lambda 1$, $\lambda 2$, $\lambda 3$, $\lambda 4$, and $\lambda 5$ be the five eigenvalues of A. Note that these eigenvalues need not be distinct. The value of $\lambda 1 + \lambda 2 + \lambda 3 + \lambda 4 + \lambda 5$ = . CS

2023 · Q.55 · 2 marks
Let G be a simple, finite, undirected graph with vertex set {v1 , . . . , vn }. Let Δ(G) denote the maximum degree of G and let N = {1, 2, . . .} denote the set of all possible colors. Color the vertices of G using the following greedy strategy: for i = 1, . . . , n color(vi) ← min{j ∈ N : no neighbour of vi is colored j} Which of the following statements is/are TRUE?
(A) This procedure results in a proper vertex coloring of G.
(B) The number of colors used is at most Δ(G) + 1.
(C) The number of colors used is at most Δ(G).
(D) The number of colors used is equal to the chromatic number of G. CS

2023 · Q.56 · 2 marks
Let U = {1, 2, 3}. Let 2U denote the powerset of U . Consider an undirected graph G whose vertex set is 2U . For any A, B ∈ 2U , (A, B) is an edge in G if and only if (i) A = B, and (ii) either A B or B A. For any vertex A in G, the set of all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by B
(A) . If ∅ denotes the empty set, then the cardinality of B(∅) is . CS

2022 · Q.30 · 1 mark
Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is _____.

2022 · Q.37 · 2 marks
Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of
(A) A3
(B) A3 divided by 2
(C) A3 divided by 3
(D) A3 divided by 6

2022 · Q.50 · 2 marks
The following simple undirected graph is referred to as the Peterson graph. Which of the following statements is/are TRUE?
(A) The chromatic number of the graph is 3.
(B) The graph has a Hamiltonian path.
(C) The following graph is isomorphic to the Peterson graph.
(D) The size of the largest independent set of the given graph is 3. (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

2022 · Q.52 · 2 marks
Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?
(A) The diagonal entries of A2 are the degrees of the vertices of the graph.
(B) If the graph is connected, then none of the entries of An 1 I n can be zero.
(C) If the sum of all the elements of A is at most 2(n 1), then the graph must be acyclic.
(D) If there is at least a 1 in each of A ’s rows and columns, then the graph must be connected.

2021 · Q.16 · 1 mark [OCR]
In an undirected connected planar graph G, there are eight vertices and five faces. The number of edges in G is

2020 · Q.52 · 2 marks [OCR]
Graph G is obtained by adding vertex s to K,, and making s adjacent to every vertex of K,,. The minimum number of colours required to edge-colour G is

2019 · Q.12 · 1 mark [OCR]
Let G be an undirected complete graph on n vertices, where n > 2. Then, the number of different Hamiltonian cycles in G is equal to
(A) n!
(B) m—1)! (1
(D) 2

2018 · Q.18 · 1 mark
The chromatic number of the following graph is _____. CS 5/20 GATE 2018

2018 · Q.43 · 2 marks
Let G be a graph with 100! vertices, with each vertex labelled by a distinct permutation of the numbers 1,2, ... , 100. There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v. Let y denote the degree of a vertex in G, and z denote the number of connected components in G. Then, y + 10z = _____. CS 15/20 GATE 2018

2017 CS1 · Q.20 · 1 mark [OCR]
Let T be a tree with 10 vertices. The sum of the degrees of all the vertices in T is =====PAGE 11=====

2017 CS2 · Q.23 · 1 mark [OCR]
Gis an undirected graph with 7 vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of 77 is =====PAGE 10=====

B. New practice questions with detailed answers

Q1 · 1 mark · NAT
The number of spanning trees of the labelled complete graph K_5 is _____.

Answer: 125
Cayley's formula: the number of labelled trees on n vertices is n^(n−2). For n = 5: 5^3 = **125**.

Q2 · 2 marks · NAT
A simple connected planar graph has 10 vertices, and every face (including the outer face) is bounded by exactly 3 edges. The number of edges is _____.

Answer: 24
Each face has 3 edges and each edge borders 2 faces, so 3F = 2E, i.e. F = 2E/3.
Euler: V − E + F = 2 → 10 − E + 2E/3 = 2 → E/3 = 8 → E = **24**.
Check: maximal planar graph has 3V − 6 = 24 edges. ✓

Combinatorics, Recurrences, Generating Functions

1.5 marks per paper · 16 questions · appeared in 12 of 15 papers. Counting, inclusion-exclusion, recurrences, generating functions.

A. Past-paper questions (16)

2026 CS1 · Q.12 · 1 mark
Consider 4 × 4 matrices with their elements from {0, 1}. The number of such matrices with even number of 1s in every row and every column is
(A) 512
(B) 1025
(C) 1023
(D) 255

2025 CS1 · Q.30 · 1 mark
Let S be the set of all ternary strings defined over the alphabet {a, b, c}. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, “aa”, “bb” or “cc”. The number of such strings of length 5 that are possible is _____. (Answer in integer)

2024 CS2 · Q.15 · 1 mark
Let T(n) be the recurrence relation defined as follows: T(0) = 1, T(1) = 2, and T(n) = 5T(n − 1) − 6T(n − 2) for n ≥ 2
Which one of the following statements is TRUE?
(A) T(n) = Θ(2n)
(B) T(n) = Θ(n2n)
(C) T(n) = Θ(3n)
(D) T(n) = Θ(n3n)

2023 · Q.15 · 1 mark
The Lucas sequence Ln is defined by the recurrence relation: Ln = Ln−1 + Ln−2 , for n ≥ 3, with L1 = 1 and L2 = 3.
Which one of the options given is TRUE? √ n √ n 1+ 5 1− 5
(A) Ln = + 2 2 √ n √ n 1+ 5 1− 5
(B) Ln = − 2 3 √ n √ n 1+ 5 1− 5
(C) Ln = + 2 3 √ n √ n 1+ 5 1− 5
(D) Ln = − 2 2 CS

2023 · Q.48 · 2 marks
Let U = {1, 2, . . . , n}, where n is a large positive integer greater than 1000. Let k be a positive integer less than n. Let A, B be subsets of U with |A| = |B| = k and A ∩ B = ∅. We say that a permutation of U separates A from B if one of the following is true. - All members of A appear in the permutation before any of the members of B. - All members of B appear in the permutation before any of the members of A. How many permutations of U separate A from B?
(A) n! n
(B) (n − 2k)! 2k n
(C) (n − 2k)! (k!)2 2k n
(D) 2 (n − 2k)! (k!)2 2k CS

2022 · Q.32 · 1 mark
The number of arrangements of six identical balls in three identical bins is_____.

2022 · Q.36 · 2 marks
Which one of the following is the closed form for the generating function of the sequence { an } n 0 defined below? n 1, n is odd an 1, otherwise
(A) x 1 x2 1 1 x 1 x 2 2
(B) x(3 x 2) 1 1 x 1 x 2 2
(C) 2x 1 1 x 2 2 1 x
(D) x 1 (1 x) 1 x 2 2

2021 · Q.19 · 1 mark [OCR]
There are 6 jobs with distinct difficulty levels, and 3 computers with distinct pro- cessing speeds. Each job is assigned to a computer such that: - The fastest computer gets the toughest job and the slowest computer gets the easiest job. - Every computer gets at least one job. The number of ways in which this can be done is CS, Set-1 - Copyright © GATE 2021

2020 · Q.42 · 2 marks [OCR]
The number of permutations of the characters in LILAC so that no character appears in its original position, if the two L’s are indistinguishable, is

2019 · Q.5 · 1 mark [OCR]
Let U = {1,2,...,n}. Let A = {(x,X)|x € X,X © U}. Consider the following two statements on |A|. L JA] = n2"-1 1 |Al= Sea k(Z) Which of the above statements is/are TRUE?
(A) Only I
(B) Only I
(C) Both I and I
(D) Neither I nor II cs Vis GATE 2019 Computer Science and Information Technology Set-2

2018 · Q.1 · 1 mark
Which one of the following is a closed form expression for the generating function of the sequence {an}, where an = 2n + 3 for all n = 0, 1, 2,... ? 3 3x 2 x 3 x
(A)
(B)
(C)
(D) (1 x)2 (1 x)2 (1 x)2 (1 x)2

2017 CS1 · Q.47 · 2 marks [OCR]
The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is

2017 CS2 · Q.47 · 2 marks [OCR]
iz @=2)°" If the ordinary generating function of a sequence {@,,}7_o is then a3 — do is equal to =====PAGE 20=====

2016 · Q.2 · 1 mark
Let an be the number of n-bit strings that do NOT contain two consecutive 1s. Which one of the following is the recurrence relation for an ?
(A) an = an−1 + 2an−2
(B) an = an−1 + an−2
(C) an = 2an−1 + an−2
(D) an = 2an−1 + 2an−2

2016 · Q.26 · 2 marks
The coefficient of x12 in (x3 + x4 + x5 + x6 + · · ·)3 is .

2016 · Q.27 · 2 marks
Consider the recurrence relation a1 = 8, an = 6n2 + 2n + an−1 . Let a99 = K × 104 . The value of K is .

B. New practice questions with detailed answers

Q1 · 1 mark · NAT
The number of binary strings of length 6 with no two consecutive 1s is _____.

> **Answer: 21**
> Let a(n) = valid strings of length n. A string ends in 0 (any valid string of length n−1 before it) or in 01 (any valid string of length n−2 before it). So a(n) = a(n−1) + a(n−2).
> a(1) = 2, a(2) = 3 → 2, 3, 5, 8, 13, **21** for n = 6 (Fibonacci).

Q2 · 2 marks · NAT
The number of integer solutions of $x_1 + x_2 + x_3 + x_4 = 20$ with every $x_i \geq 2$ and $x_1 \leq 6$ is _____.

> **Answer: 335**
> Substitute $y_i = x_i - 2 \geq 0$: $y_1 + y_2 + y_3 + y_4 = 12$ with $y_1 \leq 4$.
> All solutions: C(12+3, 3) = C(15,3) = 455.
> Bad solutions ($y_1 \geq 5$): let $y_1' = y_1 - 5$, sum = 7 → C(10,3) = 120.
> Answer = 455 − 120 = **335**. (Inclusion–exclusion / complement counting.)

Propositional & First-Order Logic

1.3 marks per paper · 14 questions · appeared in 12 of 15 papers. Translate statements to predicate logic; equivalences and validity of formulae.

A. Past-paper questions (14)

2026 CS2 · Q.11 · 1 mark
For two different persons x and y, the predicate M(x, y) denotes that x knows y. Consider the following statement. There is a person who does not know anyone else, but that person is known by everyone else. Which one of the following expressions represents the above statement?
(A) (∃y)(∀x) ((x ≠ y) → (M(x, y) ∧ ¬M(y, x)))
(B) (∀y)(∃x) ((x ≠ y) → (M(x, y) ∧ ¬M(y, x)))
(C) (∃y)(∃x) ((x ≠ y) → (M(x, y) ∧ ¬M(y, x)))
(D) (∀y)(∀x) ((x ≠ y) → (M(x, y) ∧ ¬M(y, x)))

2025 CS1 · Q.48 · 2 marks
Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: “Everyone has exactly one mother”? The meanings of the predicates used are: • mother(y, x): y is the mother of x • noteq(x, y): x and y are not equal
(A) ∀x∃y∃z(mother(y, x) ∧ ¬mother(z, x))
(B) ∀x∃y[mother(y, x) ∧ ∀z(noteq(z, y) → ¬mother(z, x))]
(C) ∀x∀y[mother(y, x) → ∃z(mother(z, x) ∧ ¬noteq(z, y))]
(D) ∀x∃y[mother(y, x) ∧ ¬∃z(noteq(z, y) ∧ mother(z, x))]

2025 CS2 · Q.15 · 1 mark
Let P(x) be an arbitrary predicate over the domain of natural numbers. Which ONE of the following statements is TRUE?
(A) (P(0) ∧ (∀x [P(x) ⇒ P(x + 1)])) ⇒ (∀x P(x))
(B) (P(0) ∧ (∀x [P(x) ⇒ P(x − 1)])) ⇒ (∀x P(x))
(C) (P(1000) ∧ (∀x [P(x) ⇒ P(x − 1)])) ⇒ (∀x P(x))
(D) (P(1000) ∧ (∀x [P(x) ⇒ P(x + 1)])) ⇒ (∀x P(x))

2024 CS2 · Q.12 · 1 mark
Let p and qbe the following propositions: p: Fail grade can be given. q: Student scores more than 50% marks. Consider the statement: “Fail grade cannot be given when student scores more than 50% marks.” Which one of the following is the CORRECT representation of the above statement in propositional logic?
(A) q → ¬p
(B) q→p
(C) p→q
(D) ¬p → q

2023 · Q.26 · 1 mark
Geetha has a conjecture about integers, which is of the form ∀x P (x) ==⇒ ∃yQ(x, y) , where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would imply Geetha’s conjecture?
(A) ∃x P (x) ∧ ∀yQ(x, y)
(B) ∀x∀yQ(x, y)
(C) ∃y∀x P (x) ==⇒ Q(x, y)
(D) ∃x P (x) ∧ ∃yQ(x, y) CS

2021 · Q.7 · 1 mark [OCR]
Let p and q be two propositions. Consider the following two formulae in proposi- tional logic. Si: (-pA(pVq))>4 Sp: q+ (-pA(pv4q)) Which one of the following choices is correct?
(A) | Both S; and S, are tautologies.
(B) | S, is a tautology but S, is not a tautology.
(C) | S, is not a tautology but S is a tautology.
(D) | Neither S, nor S is a tautology.

2020 · Q.39 · 2 marks [OCR]
Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x. (a) Vx(p(x) VW) = vx p(x) VW (8) ax(p(x) AW) = Ax p(x) AW () Vx(p(x) ~ W) = vx p(x) > W (0) ax(p(x) > W) = vx p(x) >W

2019 · Q.35 · 2 marks [OCR]
Consider the first order predicate formula g: Vx [(Wz z|x > (2 =x) V(z=1))) > AW (W>X)A(Vz z|W > ((W =z) V (z= 1)))] Here 'a|b' denotes that 'a divides b', where a and b are integers. Consider the following sets: Sl. {1,2,3, ..., 100} S2. Set of all positive integers S3. Set of all integers Which of the above sets satisfy p?
(A) Sl and $2
(B) S1 and $3
(C) $2 and $3
(D) S1, $2 and $3 cs 8/15 GATE 2019 Computer Science and Information Technology Set-2

2018 · Q.28 · 2 marks
Consider the first-order logic sentence φ ≡ ∃s∃t∃u∀v∀w∀x∀y ψ(s, t, u, v, w, x, y) where ψ(s, t, u, v, w, x, y) is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements. Which one of the following statements is necessarily true?
(A) There exists at least one model of φ with universe of size less than or equal to 3.
(B) There exists no model of φ with universe of size less than or equal to 3.
(C) There exists no model of φ with universe of size greater than 7.
(D) Every model of φ has a universe of size equal to 7. CS 8/20 GATE 2018

2017 CS1 · Q.1 · 1 mark [OCR]
The statement (— p) > (= q) is logically equivalent to which of the statements below? i. Poe Es? Il. (49) v p IV. (Ap)vq
(A) Tl only
(B) Tand IV only
(C) Il only
(D) I and III only

2017 CS1 · Q.2 · 1 mark [OCR]
Consider the first-order logic sentence F: ¥x(4yR(x.')). Assuming non-empty logical domains. which of the sentences below are implied by F? I av(rR(x.7)) I. Av(vVxR(x,¥)) I. Vy(arR(x,¥)) IV. -4x(VyAR(x,¥))
(A) IV only
(B) Iand IV only
(C) Il only
(D) I and III only =====PAGE 3=====

2017 CS1 · Q.29 · 2 marks [OCR]
Let p. g. and 7 be propositions and the expression (py — g) > 7 be a contradiction. Then. the expression ("> p)— qis
(A) a tautology.
(B) a contradiction.
(C) always TRUE when p is FALSE.
(D) always TRUE when g is TRUE.

2017 CS2 · Q.11 · 1 mark [OCR]
Let p. g. 7 denote the statements "Jt is raining". "It is cold", and "It is pleasant". respectively. Then the statement "Jr is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by (AY CPANACT > @AQ)
(B) CPA) ACAD 7) C)CPpaAnv@ag--7"
(D) CPANVE> CAD)

2016 · Q.1 · 1 mark
Let p, q, r, s represent the following propositions. p: x ∈ {8, 9, 10, 11, 12} q: x is a composite number r: x is a perfect square s: x is a prime number The integer x ≥ 2 which satisfies ¬((p ⇒ q) ∧ (¬r ∨ ¬s)) is .

B. New practice questions with detailed answers

Q1 · 1 mark · MCQ
The formula ¬∀x (P(x) → Q(x)) is logically equivalent to
(A) ∃x (P(x) ∧ ¬Q(x))
(B) ∀x (P(x) ∧ ¬Q(x))
(C) ∃x (¬P(x) ∨ Q(x))
(D) ∀x (¬P(x) ∨ ¬Q(x))

Answer: (A)
Negation flips the quantifier: ¬∀x φ ≡ ∃x ¬φ.
¬(P → Q) ≡ ¬(¬P ∨ Q) ≡ P ∧ ¬Q.
So the result is ∃x (P(x) ∧ ¬Q(x)): "some P is not Q".

Q2 · 2 marks · MSQ

Which of the following are valid (true in every interpretation)?

(A) $(p \to q) \to ((q \to r) \to (p \to r))$

(B) $((p \vee q) \wedge \neg p) \to q$

(C) $\exists x \forall y\, R(x, y) \to \forall y \exists x\, R(x, y)$

(D) $\forall y \exists x\, R(x, y) \to \exists x \forall y\, R(x, y)$

Answer: (A), (B), (C)

(A) Hypothetical syllogism — valid.

(B) Disjunctive syllogism — valid.

(C) If one x works for every y, then for each y that same x works — valid.

(D) Not valid. Counter-example: domain $\{1,2\}$, $R(x,y)$ means $x = y$. For every y some x exists (take $x = y$), but no single x equals both 1 and 2.

Rule: $\exists\forall \Rightarrow \forall\exists$ always; the converse fails.

Calculus

1.3 marks per paper · 17 questions · appeared in 15 of 15 papers. Limits, continuity/differentiability, maxima-minima, definite integrals.

A. Past-paper questions (17)

2026 CS1 · Q.32 · 1 mark
Consider the function f: R → R defined as follows: x 1 f(x) = {c1 e − c2 log e ( ) , if x > 0 x 3 otherwise where c1 , c2 ∈ R. If f is continuous at x = 0, then c1 + c2 = _____. (answer in integer)

2026 CS1 · Q.46 · 2 marks
Let f: R → R be defined as follows: |x| |x| f(x) = (− x) (x −) 2 2 Which of the following statements is/are true?
(A) f has a local maximum
(B) f has a local minimum
(C) f ′ is continuous over R
(D) f ′ is not differentiable over R

2026 CS2 · Q.27 · 1 mark
1 For a real number a, let I(a) = ∫−1(3x 2 − ax + 1) dx . Which of the following statements is/are true?
(A) The value of I(a) is independent of the value of a
(B) The value of I(a) can vary with the value of a
(C) There exists a ∈ (−∞, +∞) such that I(a) is a positive real number
(D) There exists a ∈ (−∞, +∞) such that I(a) is a negative real number

2025 CS1 · Q.31 · 1 mark
Consider the given function f(x). ax + b for x < 1 f(x) = { 3 2 x + x + 1 for x ≥ 1 If the function is differentiable everywhere, the value of b must be _____. (rounded off to one decimal place)

2025 CS2 · Q.12 · 1 mark
The value of x such that x > 1, satisfying the equation ∫1 t ln t dt = 4 is
(A) √e
(B) e
(C) e2
(D) e−1

2024 CS1 · Q.11 · 1 mark
Let f: R → R be a function such that f(x) = max{x, x 3 }, x ∈ R , where R is the set of all real numbers. The set of all points where f(x) is NOT differentiable is
(A) {−1, 1, 2}
(B) {−2, −1, 1}
(C) {0, 1}
(D) {−1, 0, 1} 1 2 3

2024 CS2 · Q.16 · 1 mark
Let f(x) be a continuous function from R to R such that f(x) = 1 − f(2 − x) 2 Which one of the following options is the CORRECT value of ∫0 f(x)dx ?
(A) 0
(B) 1
(C) 2
(D) −1

2023 · Q.28 · 1 mark
Let f (x) = x3 + 15x2 − 33x − 36 be a real-valued function. Which of the following statements is/are TRUE?
(A) f (x) does not have a local maximum.
(B) f (x) has a local maximum.
(C) f (x) does not have a local minimum.
(D) f (x) has a local minimum. CS

2023 · Q.31 · 1 mark
The value of the definite integral 3 2 1 4x2 y − z 3 dz dy dx −3 −2 −1 is . (Rounded off to the nearest integer) CS

2022 · Q.34 · 1 mark
The value of the following limit is _____. x lim x 0 1 e 2 x

2021 · Q.20 · 1 mark [OCR]
Consider the following expression. . V2r+22—4 lim ———_——_ 3-3 2+3 The value of the above expression (rounded to 2 decimal places) is

2020 · Q.1 · 1 mark [OCR]
Consider the functions Te I. x? -sinx Wi VxF 47 Which of the above functions is/are increasing everywhere in [0,1]?
(A) Ill only (8) Il only () Il and III only (0) land III only

2019 · Q.13 · 1 mark [OCR]
Cc ie lim x4-81 a x33 2x2-5x-3
(A) 1
(B) 53/12
(C) 108/7
(D) Limit does not exist

2018 · Q.16 · 1 mark
π/4 The value of ∫0 x cos(x 2)dx correct to three decimal places (assuming that π = 3.14) is _____.

2017 CS1 · Q.28 · 2 marks [OCR]
1935 The value of lim ie x>1x°—3x°42
(A) is 0
(B) is -1
(C) is 1
(D) does not exist

2017 CS2 · Q.10 · 1 mark [OCR]
If f(x) = R sin (=) +S, f' (5) = V2 and i f(x)dx = a then the constants R and S are. respectively
(A) = and =
(B) = and 0 4 4 16
(C) = and 0
(D) = and —

2016 · Q.3 · 1 mark
sin(x − 4) lim = . x→4 x−4

B. New practice questions with detailed answers

Q1 · 1 mark · NAT
$\lim_{x\to 0} (e^{2x} - 1 - 2x) / x^2$ is _____.

Answer: 2
Maclaurin series: $e^{2x} = 1 + 2x + (2x)^2/2! + \ldots = 1 + 2x + 2x^2 + \ldots$
Numerator = $2x^2$ + higher-order terms, so the ratio tends to **2**.
(L'Hôpital twice gives $4e^{2x}/2 \to 2$ as well.)

Q2 · 2 marks · MCQ
Let f(x) = |x| + |x − 2| + |x − 4|. The number of points at which f is NOT differentiable is
(A) 0 (B) 1 (C) 3 (D) infinite

Answer: (C) 3
Each |x − a| has a corner at x = a where the slope jumps from −1 to +1 (a jump of +2).
Slope of f: −3 for x < 0, −1 on (0,2), +1 on (2,4), +3 for x > 4.
The slope jumps at 0, 2 and 4, and the jumps are all positive (+2), so they cannot cancel. Not differentiable at exactly **3** points.